Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/01/2014
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Project Title:	Effects of prolonged exposure to space radiation on carcinogenesis and neuroendocrine differentiation in human prostate models		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22908-0383	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	01/01/2010	**End Date:**	12/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2013, per NSSC information (Ed., 10/10/2012) NOTE: Period of performance changed to 01/01/2010-12/31/2012 (from 11/15/09-11/14/12) per NSSC info (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Larner, James (University of Virginia) Parsons, Sarah (University of Virginia Health System)		
Grant/Contract No.:	NNX10AC13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The deleterious effects of radiation encountered in space presents a significant obstacle in our quest to explore the space. Unfortunately, due to the significant differences between space radiation and radiation encountered on Earth, the risks of cancer development following exposure to space radiation cannot be reliably predicted based solely on epidemiological data collected on Earth. This project will study the effects of space radiation on development and/or progression of prostate cancer. Prostate cancer is the most common cancer in men in the United States and the second leading cause of cancer deaths. It is also the most often diagnosed type of cancer in male astronauts. However, the connection between exposure to space radiation and the development of prostate cancer is not yet known. Several lines of evidence suggest that conversion of normal prostate or prostate cancer cells into cells secreting cell growth hormones, so called neuroendocrine cells, stimulate cancer development and/or progression. Strikingly, such conversion was recently reported in prostate cells exposed to prolonged irradiation. The low doses and long exposure times used in that study are not dissimilar to the doses and exposure times encountered during space travel. Therefore, we postulate that space radiation could increase the risk of prostate cancer development through the transformation of prostate cells into neuroendocrine cells during space flights. Since neuroendocrine cells seem to play an important role in the development of other important and common types of solid tumors, such as breast, lung, or colon, the insight gained during this study should be also applicable to these cancers. The results obtained during the proposed studies will allow NASA to better estimate the prostate cancer health risk(s) of space exploration, as well as provide clues that will allow the development of novel preventative measures to protect astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is believed that space radiation can increase the risk of tumor development and progression. We hypothesize that space radiation could not only induce neoplastic transformation in prostate cells (carcinogenesis), but in addition to direct interactions with target cells (e.g. causing carcinogenic mutations in the cellular DNA), could also stimulate surrounding cells and tissues to provide growth assistance to the developing tumor. One of the mechanisms of such stimulation could be the neuroendocrine differentiation, converting normal epithelial cells into neuroendocrine-like cells which produce hormones and growth factors. Such differentiation has been observed in prostate, breast, GI tract, and lung tissues, all important target organs for space radiation-induced cancers. Some of the research and technological questions that this proposal addresses include the following: 1) What is the relationship between induction of carcinogenesis and/or tumor progression and the dose and quality of space radiation? 2) Can the in vitro cell culture models be used as an approximation for prostate carcinogenesis and neuroendocrine differentiation in humans? 3) How can effective biomarkers of carcinogenic and neuroendocrine differentiation risk from space radiation be developed and validated? 4) What is the likelihood that neuroendocrine differentiation will increase the risk of carcinogenesis in prostate and other target organs? Better understanding of the mechanisms underlying the effects of exposure to radiation, including the role of stimulating activities of cancer and surrounding cells, will provide significant advantages in predicting cancer risk from space and/or terrestrial radiations.
Task Progress:	Our results demonstrate that cellular responses to space-like ionizing radiation in prostate cells depend on radiation quality (e.g. LET) and dose. Importantly, the responses also differ depending on the cell type (normal vs. early cancer cells). Dose- and LET-dependence is clearly visible in both cell survival and carcinogenic transformation, with higher LET oxygen and titanium ions causing significantly bigger effects than low LET energetic protons and X-ray photons. In addition, we have confirmed the crucial role of neurotensin signaling and neurotensin receptor 1 (NTR1) expression in prostate cancer cell survival and radioresistance. Most importantly, following irradiation we observed induction of NTR1 protein expression in normal prostate epithelial cells and increased NTR1 expression in prostate cancer cells. Based on these observations we proposed that NTR1 expression, and/or neurotensin signaling, enhances resistance to ionizing radiation exposure and provides additional pro-survival stimuli in irradiated prostate cells. Our study demonstrated that inhibition, or downregulation of the NTR1 receptor reduces cell survival and enhances apoptotic cell death following irradiation of prostate cancer cells. Moreover, the molecular mechanism(s) of this sensitization involves EGFR receptor signaling. This role of EGFR receptor in the cellular responses to ionizing radiation and its importance for DNA damage repair has been also proposed in the literature We postulate that increased NTR1 expression in irradiated normal prostate epithelial cells not only coincides with carcinogenic transformation but could be involved in the process. One of the possible mechanisms of such involvement could be a non-physiological stimulation of androgen receptor (AR), resulting in enhancement of its transcriptional and DNA binding ability. It has been shown previously that activated AR may participate in creating genomic rearrangements during NHEJ-mediated DNA double-strand breaks repair. The carcinogenic rearrangements observed in this process include TMPRSS2-ETS translocation often associated with prostate cancer. Importantly for this project, we have demonstrated increased expression of ERG1/2/3 (members of ETS family) in normal prostate cells exposed to space-like radiation. In the following year we will focus on determining the molecular mechanism of carcinogenic transformation in irradiated prostate cells, including possibility of induction of TMPRSS2-ETS translocations. Unlike carcinogenic transformation detected following exposure to single doses of radiation, we have not detected any markers (morphological changes or protein expression/secretion) of neuroendocrine differentiation in irradiated prostate cells. Based on these results, we conclude that a single-dose exposure, even using high LET, high RBE space-like radiation is not sufficient to induce neuroendocrine differentiation in prostate cells. Altogether, the data obtained in this NASA funded project demonstrate that radiation dose and quality have clear differential effects not just on DNA damage and cell killing, but more importantly on the initiation of prostate carcinogenesis. While it is not possible to irradiate cells at the low fluences that would be experienced by astronauts during space exploration, our data also demonstrate that fractionated irradiations, but not a single acute exposure, induce mesenchymal-to-epithelial transition or neuroendocrine differentiation of normal prostate cells and prostate cancer cells. Further, our work defines signaling pathways and genetic alterations that effect these molecular and phenotypic changes that are involved in radiation-induced prostate cancer initiation and progression. As such, this project increases the mechanistic understanding of the fundamental biological processes involved in space radiation-induced carcinogenesis and provide NASA with important information regarding the potential increase risk of prostate cancer as a result of space radiation exposure.

Bibliography Type:	Description: (Last Updated: 11/14/2018)